INCH-POUND

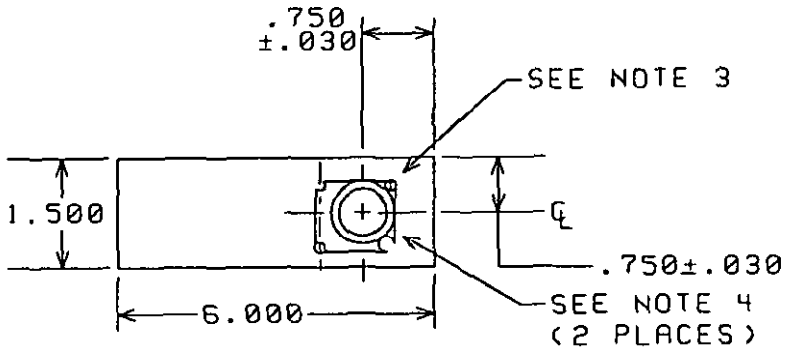
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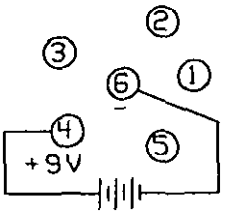
PERFORMANCE SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, HIGH PERFORMANCE, BA-X599/U

This specification sheet is approved for use within Army Research Laboratory, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-PRF-49471(ER).





TERMINAL MARKING



SIDE VIEW

Figure 1. Battery dimensions.

AMSC N/A FSC 6135

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NOTES:

1. All dimensions are in inches.

2. Tolerances, unless otherwise specified, are -.063, +.000.

3. The battery terminal shall consist of a miniature circular battery connector, Dwg SC-C-179492 or equal. Connector shall be mounted in accordance with referenced drawing.

4. Connector socket body configuration shall accept two (2) each 0.156 inch diameter guideposts as shown. Depth of cavity shall be 0.657 inch minimum.

5. The socket shall be supported and mounted so that the top surface of the socket shall not protrude above the adjacent outside surface of the jacket and shall not be more than 1/16 inch below adjacent outside surface of the jacket when used initially, during, and after subsequent insertions of the mating plug during contractual testing.

REQUIREMENTS:

Dimensions and configuration: See Figure 1.

Maximum voltage: 9.15 volts.

Minimum voltage for capacity: 6.00 volts

Charge protection device: Applicable.

Safety feature: Applicable.

Terminals: Six (6) hole socket type with no obstructions of any holes (see note 5).

Weight (maximum): 1.0 pound (454 grams).

Capacity tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

Capacity Test	Minimum Capacity Requirements in Hours to 6.00 Volts
I	3.2
L	1.7
H	6.2
HT	6.0
LT	1.6
IT	3.0

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Initial voltage delay: During the I, H, HT, and IT discharge tests, initial voltages below 6.0 volts cannot exceed a 5.0 second duration. During the L and LT discharge tests initial voltages below 5.70 volts cannot exceed a 60.0 second duration.

High temperature device: Applicable.

Complete discharge device: Applicable.

Connector: The connector shall be in accordance with SC-C-179492 or equivalent with the following exception: The durability test described in paragraph 4h shall be performed 50 times with SM-D-687888 or equivalent. Each of the metallic sockets in the connector (top view) shall conform to Figure 2. Each socket shall be cylindrical in shape and may have a single gap with a maximum opening of 30 degrees. The socket shall accept a number 16 pin.

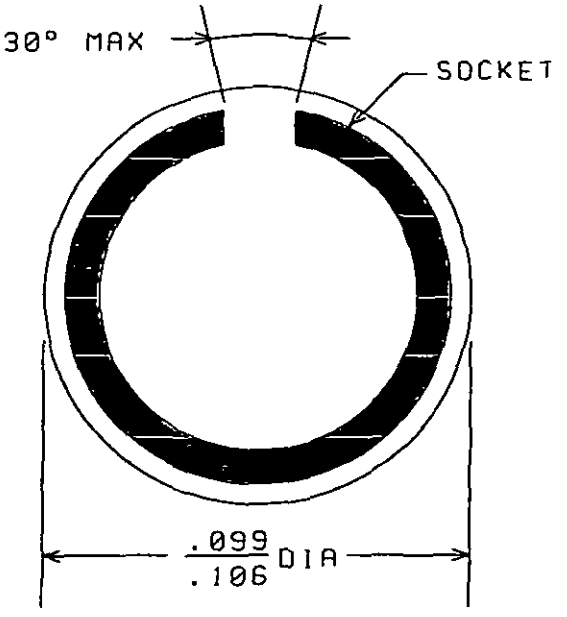


Figure 2. Socket (Top View)

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METHODS OF EXAMINATION AND TESTS:

Capacity tests:

(1) Storage:

Details on storage conditions for all capacity tests are specified in the basic specification.

(2) Discharge:

Cells (forced discharge): Two cells shall be discharged at 2.0 amperes at 70±5°F to 2/3 of open-circuit voltage. Each discharged cell shall then be connected in series to two fresh cells. Each string shall be discharged at 2.0 amperes at 70±5°F to 2/3 open-circuit voltage of the two fresh cells.

Batteries: For all capacity tests, except the H and HT, the batteries shall be discharged at two amperes constant current to zero volts. For the H and HT capacity tests the batteries shall be discharged through two amperes constant current for five minutes followed by open circuit for five minutes. This cycle shall be repeated until the closed-circuit voltage reaches zero volts.

Closed-circuit voltage test: Load resistance of 8 ohms shall be used. Voltage shall be above 6.00 volts within 5.0 seconds.

Custodian:
Army-ER

Preparing Activity:
Army-ER

(Project 6135-A274-01)

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